

Question:
At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<tool_call> reasoning process here

At birth , the brain undergoes rapid development , but different regions develop at different times . The brain stem , which controls vital functions like breathing and heart rate , is one of the first structures to develop and is present very early in embryonic development . In contrast , the cortex , especially the prefrontal cortex (a part of the frontal lobe) , is the most recently developed region . The cortex is responsible for higher cognitive functions such as decision-making , planning , and reasoning , and it develops gradually over the first few years of life .

Therefore , the least developed part of the brain at birth is the cortex , particularly the prefrontal cortex , which takes years to mature . The frontal lobe , which is part of the cortex , is the last of the major brain areas to fully develop .

While other structures like the limbic system or basal ganglia have earlier developmental milestones , the most developed in infancy , but the cortex (especially its outer layers) remains underdeveloped at birth .

So the correct answer is the one reflecting the part of the brain that is the last to develop and thus considered the least developed at birth âĖĶ which is the cortex .

Answer choice G is "cortex" which is correct .

Among the options , the cortex is the key structure that is still immature at birth and develops over months and years .

Thus , the least developed part of the brain at birth is the cortex .

<answer> G </answer>

